

Guesstimation and Measurements of Length, Mass, and Time

Synopsis: Length in meters (m), mass in kilograms (kg), and time in seconds (s) are three fundamental physical quantities that combine to form other derived quantities. For example, velocity is length/time (m/s), acceleration is velocity/time (m/s/s = m/s²), force is mass-acceleration (kg·m/s²), and energy (J) is force-distance (kg·m/s²·m = kg m²/s² = J). In this lab, we aim to get a feel for how long, massive, and fast the things we typically experience in everyday life are. First, we will try to guess in meters, kilograms, and seconds (MKS units). Then, we will compare the measurements using a ruler, balance, and stopwatch. Physicists are often guided by an intuitive "feel" for the sizes of quantities to check calculations and consistency in scientific reasoning.

Performance Expectations: Know the rough ranges of lengths and masses of typical objects and the time durations of events around us.

Guesstimate lengths and masses of objects and the time duration of events. Then, compare them to measurements with a tape rule, balance, and stopwatch. Calculate the % error using the formula:

$$\% \text{ error} = \frac{|Guessed - Measured|}{Measured} \times 100$$

Answer questions following the error calculation.

I. Lengths

Length	Guessed (m)	Measured (m)	% error
Length of workbench			
Height of workbench			
Width of classroom door			
Length of classroom (F to B)			
Height of ceiling			
Width of your stride			
Dr. Cho's height			

II. Masses

Masses	Guessed (kg)	Measured (kg)	% error
Cell phone			
Wallet			
Watch			
Physics textbook			
Laptop			
Pilot G-2 pen			
My mass			

III. Time

Time duration	Guessed (s)	Measured (s)	% error
Rubber ball dropped from workbench edge			
Rubber ball dropped from ceiling			
Metal ball dropped from ceiling			
Coin dropped from ceiling			
Walk across classroom at normal walking (F to B)			

Questions:

- 1) You want to know how far the elevator is from the physics classroom on the second floor. Using your average stride length, how would you determine the length of a chosen path from here to the elevator?
- 2) You have determined the path's length and now want to know how long it would take to get to the elevator at a normal walking speed. But first, you wonder what your normal walking speed is. Using your measurement of classroom length and the time to walk across the classroom, determine your walking speed.
- 3) Using the walking speed you determined above, predict how long it would take to walk to the elevator. Then, time yourself walking to the elevator to see how closely your prediction matches the measured time. Show the % error.